LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Slip gauges introduction, Indian standards slip gauges, care and use of slip gauges
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Slip gauges introduction, Indian standards slip gauges, care and use of slip gauges
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of slip gauge handling, cleaning, and storage; Safety: Handle with tissue, avoid dropping; Supervision: Instructor monitors technique
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of slip gauges in metrology?
(b) Name two common materials used to manufacture slip gauges.
(c) Why must slip gauges be handled with clean hands or tissue?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Engineering Metrology • Definition and function of slip gauges • Historical context of gauge blocks • Basic terminology and nomenclature • Importance of precision in measurements	L & Ds	PPT, IB	15 min
(b)	• Indian Standards for slip gauges • Material composition and properties • Grades and tolerances (Grade 0 to 3) • Surface finish and flatness requirements • Standard sizes and series available	L & Ds	PPT, PDF	20 min
(c)	• Proper handling techniques • Cleaning and maintenance procedures • Storage methods and environmental control • Common errors and damage prevention • Inspection and verification practices	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between Grade 0 and Grade 3 slip gauges in terms of accuracy.
(b) Describe the correct method for wiping a slip gauge before use.
(c) Why is it critical to store slip gauges in a controlled environment?

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of slip gauge usage apply to the upcoming lesson on dial indicators (LP 07), focusing on precision measurement continuity.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Dial indicators advantages, classifications, care and use of dial indicators	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min